

Data Structures

Mumbai University Engineering Syllabus - Pre-loaded Reference for Conscious Commute.

Module 1: Introduction to Data Structures & Arrays

Introduction, analysis of algorithms, linear and non-linear data structures, asymptotic notations. Arrays: multi-dimensional, representation, operations, application to sparse matrices.

Module 2: Stacks and Queues

Stacks: ADT, implementation, applications (infix-to-postfix conversion, evaluation of expression).
Queues: ADT, circular queue, double-ended queue, priority queue, application.

Module 3: Linked Lists

Singly linked lists, Doubly linked lists, Circular linked lists, operations, applications (polynomial addition, sparse matrices).

Module 4: Trees

Binary trees, representation, traversal algorithms. Binary Search Trees (BST): operations, AVL trees, B-trees and B+ trees.

Module 5: Graphs

Representation (adjacency matrix, adjacency list), traversal (BFS, DFS), Minimum Spanning Trees (Kruskal's, Prim's), Single Source Shortest Path (Dijkstra's).

Module 6: Sorting, Searching & Hashing

Sorting: Bubble, Selection, Insertion, Quick, Merge, Radix. Searching: Linear and Binary search.
Hashing: Hash functions, collision resolution techniques.